

José Nicolás Rosas

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CONTACT INFORMATION

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Graduate Coordinator:	Laura Agustí Roig	laura.agusti@upf.edu	(+34) 93 542 1863

EDUCATION

2021 - present	Universitat Pompeu Fabra , Ph.D. in Economics, Barcelona, Spain
2020 - 2021	Universitat Pompeu Fabra , MRes in Economics, Finance and Management, Barcelona, Spain
2019 - 2020	Barcelona School of Economics , Master Program in Economics, Barcelona, Spain
2014 - 2016	Universidad de los Andes , Master of Arts in Economics, Bogotá, Colombia
2010 - 2014	Universidad de los Andes , Bachelor of Arts in Business Administration, Bogotá, Colombia
2008 - 2014	Universidad de los Andes , Bachelor of Arts in Economics, Bogotá, Colombia

REFERENCES

Professor **Jordi Galí** (*Advisor*)
CREi, UPF, and BSE
Tel. (+34) 93 542 2754
jgali@crei.cat

Professor **Marc Hofstetter**
Universidad de los Andes
Tel. (+57) 601 332 4493 ext. 3633
mahofste@uniandes.edu.co

Professor **Davide Debortoli** (*Advisor*)
UPF, CREi, and BSE
Tel. (+34) 93 542 1657
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Professor **Priit Jeenas**
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JOB MARKET PAPER

Carbon Emissions and the Transmission of Monetary Policy

Abstract: This paper studies the dynamic causal effects of monetary policy on carbon emissions in the United States. Using high-frequency changes in financial asset prices around Federal Open Market Committee (FOMC) announcements (i.e. monetary policy surprises), I identify a structural monetary policy shock. An analysis of the effects of these shocks reveals that, contrary to conventional wisdom, a contractionary monetary policy shock is associated with a one percent *increase* in total carbon emissions from energy consumption: while emissions from the industrial sector decline (as expected), emissions from non-industrial sectors rise significantly in the short run. A detailed exploration reveals that the channels of monetary policy transmission vary in strength and relevance across sectors and help explain these heterogeneous responses: while the conventional *aggregate demand* channel plays a central role in the response of industrial sector emissions, the evidence suggests a more significant role of *commodity price* and *energy reallocation* channels for the transmission of shocks to non-industrial sectors.

PUBLICATIONS

The Poor and the Rich: Preferences Over Inflation and Unemployment with Marc Hofstetter, June 2021, *Journal of Money, Credit and Banking*, Volume 53, Issue 4, pages 875-895.

Abstract: What are the tradeoffs that the public is willing to accept between inflation and unemployment? We find that people dislike extra points of unemployment more than extra points of inflation. This is true for both Europe and Latin America. For the latter, the aversion to unemployment relative to inflation is much greater. Moreover, in both regions, the poor's distaste for extra points of unemployment relative to inflation is significantly greater than that of the rich. This result contributes to the literature on the costs of inflation and questions the commonly held view that prescribes strong anti-inflationary postures to implement policies consistent with the preferences of the poor.

Ponzi schemes and the financial sector: DMG and DRFE in Colombia with Marc Hofstetter, Daniel Mejía and Miguel Urrutia[†], November 2018, *Journal of Banking & Finance*, Volume 96, pages 18-33.

Abstract: We use a novel dataset to estimate, for the first time in the literature, the effects of Ponzi schemes on the formal financial sector. DMG and DRFE, two Ponzi schemes that were shut down by the Colombian government in November 2008, had over half a million customers, who invested funds corresponding to 1.2% of Colombia's annual GDP. We find that pyramid customers obtained more loans from the financial sector and their credit standings were better than those in the respective control groups while the schemes were operating. Afterwards, their loan stocks started to decrease and their ratings with the banking sector deteriorated. Prior to November 2008, deposits in the financial sector fell more in the municipalities more affected by the schemes.

POLICY PUBLICATIONS

The African Continental Free Trade Area: Potential Economic Impact and Challenges with Lisandro Abrego, Mario de Zamaroczy, Tunc Gursoy, Garth P. Nicholls and Hector Perez-Saiz, May 2020, *IMF Staff Discussion Notes No. 2020/04*, International Monetary Fund.

Abstract: Political momentum towards Africa-wide free trade has been intensifying. In March 2018, over 40 countries signed the African Continental Free Trade Area (AfCFTA) agreement. Once fully implemented, the AfCFTA is expected to cover all 55 African countries, with a combined GDP of about US\$2.2 trillion. This SDN takes stock of recent trade developments in Sub-Saharan Africa and assesses the potential benefits and costs of the AfCFTA, as well as challenges to its successful implementation. In addition to increased trade flows both in existing and new products, the AfCFTA has the potential to generate substantial economic benefits for African countries. These benefits include higher income arising from increased efficiency and productivity from improved resource allocation, higher cross-border investment flows, and technology transfers. Besides lowering import tariffs, to ensure these benefits, African countries will need reduce other trade barriers by making more efficient their customs procedures, reducing their wide infrastructure gaps, and improving their business climates. At the same time, policy measures should be taken to mitigate the differential impact of trade liberalization on certain groups as resources are reallocated in the economy and activities migrate to locations with comparatively lower costs.

Trade Potential in Southern Cone Countries with Soledad Feal-Zubimendi, Fabiano Rodrigues Bastos and Daniel Hernaiz, March 2018, *IDB Technical Notes No. IDB-TN-01397*, Inter-American Development Bank.

Abstract: A trade gravity model is used to calculate the bilateral export potential of Southern Cone countries (Argentina, Brazil, Chile, Paraguay, and Uruguay) and identify patterns of weak performance across destinations and productive sectors. After controlling for key trade fundamentals and country and time effects, we find that exports to large, advanced economies like the United States and the European Union fall short of expected levels. Potential trade with the rest of Latin America shows a high degree of heterogeneity across sectors and countries. Exports within the Southern Cone are generally above the predicted value, but there is still space for higher flows, particularly regarding Chile-MERCOSUR relations.

WORKING PAPERS

The Transmission of Climate Shocks: The Case of Floods in India with Alejandro Rábano

Abstract: This paper studies the propagation of natural disasters through input-output linkages. We combine manufacturing data from the Annual Survey of Industries in India for 2000-2007 with the flood archive from the Dartmouth Flood Observatory and apply the empirical strategy suggested by de Chaisemartin and d'Haultfoeuille (2022). Output and capital decrease after the first flood hits a district, especially in areas with the lowest historical exposure. This suggests that adaptation plays a role in mediating the impact of extreme weather events. We also find that manufacturing industries that rely on agricultural inputs suffer a larger decrease in output and capital, and their output price increases, which provides evidence of a supply shock with persistent impacts. At the product level, we find some evidence of propagation through input-output linkages: when a high share of the production of a specific product is located in areas affected by extreme floods, establishments manufacturing the same type of product, but not directly affected, experience an increase in the price and a decrease in the quantity produced.

Lessons from a Bubble Burst with Marc Hofstetter

Abstract: In 2008 two Ponzi schemes were shut down by the Colombian government. Using microdata for a sample of almost a quarter of a million of their investors we (i) describe their socioeconomic characteristics and (ii) estimate the individual and household risk factors associated with making a profit from such scams. Our paper contributes to the household finances, financial education, and financial literacy literatures.

WORK IN PROGRESS

The Rise of Market Power and the Flattening of the Phillips Curve

Abstract: In recent decades, the U.S. economy has exhibited two notable trends: (i) a steady increase in aggregate market power, measured through firm-level markups and profitability, and (ii) a flattening of the Phillips curve, which indicates a weakened

relationship between inflation and real activity over the business cycle. However, the connection between these two trends is at odds with the predictions of the standard New Keynesian model. In this paper, I use a multi-sector model with production networks, incorporating heterogeneity in input-output linkages, the degree of nominal rigidities, and the degree of market power to reconcile these phenomena. Changes in sectoral markups affect the optimal demand for inputs across sectors and, consequently, the centrality of those sectors in the economy. The model predicts a New Keynesian Phillips curve that is a modified version of the one-sector model, where the characteristics of the network, sectoral price rigidities, and sectoral markups influence inflation dynamics. Quantitatively, variations in these factors can explain the flattening of the Phillips curve, with the largest effect coming from the increase in average market power.

FIELDS OF INTEREST

<i>Primary</i>	Macroeconomics
<i>Secondary</i>	Monetary Economics, Time Series Econometrics, Climate and Energy

TEACHING EXPERIENCE

Universitat Pompeu Fabra	<i>Barcelona, Spain</i>
Teaching Assistant (Undergraduate level)	
International Economics - Professor Martí Mestieri	2022
Probability and Statistics - Professor Elisa Alòs	2022
Data Analysis - Professor Claudia Sanguinetti	2021
Universidad de los Andes	<i>Bogotá, Colombia</i>
Teaching Assistant (Undergraduate level)	
Introduction to Macroeconomics - Professor Marc Hoftstetter	2014 - 2016
Intermediate Microeconomics - Professor Ana María Ibañez	2014 - 2016
Intermediate Macroeconomics - Professor Daniel Mejía	2014 - 2015
Monetary Policy and Theory - Professor Marc Hoftstetter	2010 - 2014

RESEARCH AND WORK EXPERIENCE

Universitat Pompeu Fabra	<i>Barcelona, Spain</i>
Research Assistant for Professor Andrea Caggese	04/2022 - 02/2023
International Monetary Fund	<i>Washington, DC, United States</i>
Research Analyst at the African Department (AFR-C2)	09/2018 - 08/2019
Inter-American Development Bank	<i>Washington, DC, United States</i>
Economics Contractual - Research Fellow at the Southern Cone Department	08/2016 - 07/2018
Universidad de los Andes	<i>Bogotá, Colombia</i>
Research Assistant for Professors Marc Hofstetter and Daniel Mejía	06/2014– 07/2016

HONORS, AWARDS AND GRANTS

Honors and awards	
2024	Best Paper by a Junior Researcher at the 2nd Edition of the UEA ECO Time Series Workshop
Fellowships and grants	
2022	Agència de Gestió d'Ajuts Universitaris i de Recerca (AGAUR), Grant for the recruitment of early-stage research staff (FI-2022)
2021	Universitat Pompeu Fabra, Teaching assistance fellowship
2020	Universitat Pompeu Fabra, Full MRes tuition waiver

CONFERENCES (* scheduled)

2024	2024 European Winter Meeting of the Econometric Society (Palma de Mallorca)*, RES PhD Conference 2024 (Portsmouth)*, 7th Annual Meeting of CEBRA's International Finance and Macroeconomics Program (Luxembourg)*, National Academies' Workshop on the Macroeconomic Implications of Decarbonization Policies and Actions (online), Econometric Models of Climate Change (EMCC) VIII (Cambridge), 6th QMUL
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Economics and Finance Workshop for PhD & Post-doctoral Students (London), 2nd Edition of the UEA ECO Time Series Workshop (Norwich), BSE Ph.D. Jamboree 2024 (Barcelona), CREi Macroeconomics Lunch Workshop (Barcelona)

2023 BSE Ph.D. Jamboree 2023 (Barcelona), CREi Macroeconomics Lunch Workshop (Barcelona)

LANGUAGES AND TECHNICAL SKILLS

Software & Programming Proficient in Stata, Matlab, EViews, Gretl, Dynare, LaTeX, Bloomberg, Refinitiv Datastream. Familiarity with R, Python, QGIS.

Languages Spanish (Native), English (Fluent), Portuguese (Advanced), German (Basic), Catalan (Basic).

Nationality Colombian, Spanish